

$$\begin{aligned}
& \frac{1 + pi}{5 + 3i} \\
&= \frac{(1 + pi)(5 - 3i)}{34} \\
&= \frac{5 - 3i + 5pi + 3p}{34} \\
& \frac{5 + 3p}{34} = \frac{5p - 3}{34} \\
& p = 4
\end{aligned}$$

(e) Prove that the quadratic equation  $x^2 + 3kx + k^2 = 7x + 3k$  will have two distinct real solutions for all real values of  $k$ .